IntSeqBERT: Dual-Stream Masked Modelling for Integer Sequences via Magnitude and Modular Arithmetic Embeddings

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Abstract. Integer sequences in the OEIS span values from single-digit constants to astronomical factorials and exponentials, making prediction challenging for standard tokenised models that cannot handle out-of-vocabulary values or exploit periodic arithmetic structure. We present **IntSeqBERT**, a dual-stream Transformer encoder for masked integer-sequence modelling on OEIS. Each sequence element is encoded along two complementary axes: a continuous log-scale Magnitude embedding and Sin/Cos Modulo embeddings for 100 residues (moduli 2–101), fused via FiLM. Three prediction heads (magnitude regression, sign classification, and Modulo prediction for 100 moduli) are trained jointly on 274,705 OEIS sequences. At the Large scale (91.5M parameters), IntSeqBERT achieves 95.85% Magnitude accuracy and 50.38% Mean Modulo Accuracy (MMA) on the test set, outperforming a standard tokenised Transformer baseline by +8.9 pt and +4.5 pt, respectively. An ablation removing the Modulo stream confirms it accounts for +15.2 pt of the MMA gain and contributes an additional +6.2 pt to Magnitude accuracy. A CRT-based IntegerSolver converts the model's predictions into concrete integers, yielding a 7.4× improvement in next-term prediction (Top-1: 19.09% vs. 2.59%). Modulo spectrum analysis reveals a strong negative correlation between Normalised Information Gain (NIG) and Euler's totient ratio $\varphi(m)/m$ ($r = -0.851$, $p < 10^{-28}$), providing empirical evidence that composite moduli capture OEIS arithmetic structure more efficiently via CRT aggregation.

Keywords: Integer sequences · OEIS · Masked sequence modelling · Modular arithmetic · Transformer · FiLM

1 Introduction

The On-Line Encyclopedia of Integer Sequences (OEIS) [14], as of January 2026, catalogues more than 391,000 entries spanning combinatorics, number theory, algebra, and many other branches of mathematics, making it the *de facto* standard reference for integer sequences. Each entry associates a finite integer sequence with its mathematical definition, rendering the OEIS a uniquely machine-readable corpus of mathematical knowledge.

The task we formalise is *masked sequence modelling*: a random subset of positions in a sequence is masked and the model is trained to predict each masked value from its surrounding context. This task directly probes how well a model has internalised the arithmetic and combinatorial laws governing integer sequences; next-term prediction is treated as a special case and evaluated via a dedicated Solver component. The prior benchmark FACT [1] defines and evaluates six tasks including unmasking, establishing what a plain Transformer can achieve, but it faces fundamental limitations in handling large integer values unseen at training time and in learning multiplicative structure.

The principal difficulty of this task is the extreme heterogeneity of integer sequences. Values range from single-digit constants to astronomically large factorials and exponential sequences, with differences of tens of orders of magnitude within a single training corpus. At the same time, many sequences obey *residue constraints*—parity, combinatorial patterns, or periodic remainders—that are independent of the magnitude of individual values. The standard tokenisation approach of assigning each integer a discrete vocabulary token is fundamentally ill-suited to this setting: it cannot handle large integers unseen at training time, embeds arithmetic structure in opaque token IDs, and breaks down in scale for large numbers.

We propose **IntSeqBERT**, a Transformer encoder pre-trained on the OEIS corpus via masked sequence modelling, which overcomes the above challenges through a *dual-stream* input representation. Rather than tokenising integers, IntSeqBERT encodes each element along two complementary axes:

- **Magnitude stream**: a continuous logscale embedding of the absolute value, capturing growth behaviour and scale.
- **Modulo stream**: Sin/Cos embeddings for 100 residues modulo 2 through 101, capturing periodicity and number-theoretic structure.

The two streams are fused via FiLM (Feature-wise Linear Modulation) [10]. During training, three predictors are jointly optimised in a multi-task objective: magnitude regression, sign classification, and Modulo prediction for 100 moduli. Concrete integer values are recovered from the masked-position predictions (magnitude, sign, residue distributions) using a Chinese Remainder Theorem (CRT) based **IntegerSolver** (Section 3.9).

We evaluate IntSeqBERT against two baselines on a dataset of 274,705 OEIS sequences across three model sizes (Small: $\sim$9M, Middle: $\sim$44M, Large: $\sim$110M parameters). All experiments are conducted on a single GeForce RTX 3070 Ti (8 GB VRAM), and results should be interpreted in light of this consumer-GPU memory constraint. Our main findings are:

- Large-scale IntSeqBERT achieves **95.85%** Magnitude accuracy and **50.38%** MMA on the test set (+8.9 pt and +4.5 pt over Vanilla).
- The dual-stream representation yields a **7.4**× improvement in exact-match next-term prediction (Solver Top-1: 19.09% vs. 2.59%).
- Modulo spectrum analysis reveals NIG is strongly negatively correlated with $\varphi(m)/m$ ($r = -0.851$, $p < 10^{-28}$): composite moduli aggregate arithmetic structure more efficiently via CRT.

– Modulo and Solver accuracy scale more steeply with model size than Magnitude accuracy.

Contributions.

1. *IntSeqBERT*: a dual-stream Transformer architecture fusing magnitude and modular-arithmetic feature embeddings via FiLM, jointly trained on OEIS with magnitude regression, sign classification, and 100-dimensional Modulo prediction.
2. *Comprehensive evaluation*: scale-stratified Magnitude analysis, Modulo spectrum analysis, and Solver-based next-term prediction demonstrating that composite moduli capture periodic arithmetic structure more efficiently (NIG vs. $\varphi(m)/m$: $r = -0.851$).

Paper organisation. Sections 2–4 review related work, present the architecture, and describe the experimental setup. Section 5 reports results, Section 6 provides analyses, and Section 7 concludes.

2 Background and Related Work

2.1 AI for Mathematics

Throughout the history of mathematics, numerical pattern observation has seeded important conjectures—from the Monstrous Moonshine conjecture [4] and the BSD conjecture [2] to Candelas et al.'s mirror-symmetry prediction of 317,206,375 rational curves of degree 3 [3]. Recent AI systems—AlphaGeometry [15], AlphaProof [9], the Ramanujan Machine [11], and FunSearch [13]—demonstrate the viability of AI-driven mathematical discovery. Automatically extracting arithmetic laws from integer sequence patterns at scale remains, however, largely unexplored; this work aims to establish representational foundations for that capability.

2.2 AI Research on OEIS

Program-synthesis approaches to OEIS include Alien Coding [6], which searches for the shortest reproducing code; QSynt [7], which synthesises programs for 43,516 sequences via self-learning tree search; and Learning Conjecturing [8], which automatically proves 5,565 arithmetic problems. FACT [1] (Belčák et al., NeurIPS 2022) constructs a comprehensive OEIS benchmark covering six tasks—including *unmasking*, identified as the hardest. Our Vanilla baseline corresponds to FACT's plain Transformer; IntSeqBERT extends it with Modulo feature engineering and FiLM fusion to overcome FACT's limitations with large integers and multiplicative structure.

2.3 Deep Learning and Arithmetic: Spectral Bias

Deep networks exhibit a well-known *spectral bias* [12]: they learn low-frequency functions before high-frequency ones. In the context of integer sequences, this is particularly problematic for multiplicatively growing sequences (e.g., n^2, $n!$), where learning the growth law from token sequences alone requires many layers. This work mitigates the problem through feature engineering: because the results of multiplication appear naturally in modular residues, supplying the Modulo spectrum as explicit input features reduces the network depth required to "rediscover" multiplicative structure.

2.4 Why Modulo Spectra? Number-Theoretic Motivation

Most OEIS sequences are generated by finite algorithms combining addition, multiplication, and modular operations. Crucially, modular arithmetic is a *ring homomorphism*:

$$(a+b) \bmod m = \big((a \bmod m) + (b \bmod m)\big) \bmod m, \tag{1}$$

$$(a \times b) \bmod m = \big((a \bmod m) \times (b \bmod m)\big) \bmod m. \tag{2}$$

Hence, the residue sequence $(x_i \bmod m)$ faithfully preserves the additive and multiplicative structure of the original sequence as a compact representation, regardless of the magnitudes involved.

Fermat's little theorem implies that $(a^n \bmod p)$ is periodic with period dividing $p-1$ for prime p with $\gcd(a,p) = 1$. Gauss's law of quadratic reciprocity [5] and its generalisations further link residue information across different primes. Composite moduli simultaneously retain information from all prime power factors via the Chinese Remainder Theorem (CRT), motivating the use of both prime and composite moduli in our spectrum. Our Modulo features cover $m \in \{2, 3, \ldots, 101\}$ (100 moduli), capturing power-residue structure broadly across primes and composites.

2.5 Neural Representations of Numbers

Tokenisation (the LLM mainstream) cannot handle out-of-vocabulary values and buries arithmetic structure in opaque token IDs. Log-scale continuous embeddings improve scale invariance but miss arithmetic periodicity. We instead repurpose sinusoidal embeddings [16] from positions to *modular residues of values*, and apply FiLM [10] to modulate the Magnitude stream via the Modulo stream.

2.6 Positioning of This Work

In contrast to the Urban group's program-synthesis line [6,7,8], which explicitly describes each sequence's generation rule, this work uses *representation learning* to acquire shared arithmetic structure across the OEIS corpus. Where FACT [1] identified the limits of a plain tokenised Transformer, IntSeqBERT overcomes them via Modulo-spectrum feature engineering and FiLM fusion.

3 IntSeqBERT

3.1 Problem Formulation

Let $\mathbf{x} = (x_1, x_2, \ldots, x_L)$ with $x_i \in \mathbb{Z}$ and $L \leq 128$ be a finite prefix of an OEIS sequence. We adopt *masked sequence modelling*: a random subset of positions is masked and the model is trained to predict each masked value. Specifically, for each masked position i, three quantities are predicted:

1. **Magnitude**: $v_i = \begin{cases} 0 & (x_i = 0), \\ 1 + \log_{10}|x_i| & (x_i \neq 0) \end{cases} \in \mathbb{R}_{\geq 0}$
2. **Sign**: $s_i \in \{+, -, 0\}$ (3-class label)
3. **Residues**: $r_i^{(m)} = x_i \bmod m$ for each $m \in \{2, 3, \ldots, 101\}$ (100 independent classification targets)

This decomposition separates magnitude, sign, and periodic arithmetic structure into complementary supervision signals.

3.2 Input Feature Extraction

Two feature vectors are computed for each element x_i prior to learnable embeddings.

Magnitude features $\mathbf{f}_i^{\text{mag}} \in \mathbb{R}^4$:

$$\mathbf{f}_i^{\text{mag}} = \big[v_i,\ \mathbf{1}[x_i > 0],\ \mathbf{1}[x_i < 0],\ \mathbf{1}[x_i = 0]\big].$$

The last three components are a one-hot sign representation. For astronomically large integers exceeding the `float64` range, $|x_i|$ is replaced by its decimal digit count.

Modulo features $\mathbf{f}_i^{\text{mod}} \in \mathbb{R}^{200}$: For each modulus $m \in \{2, \ldots, 101\}$, let $r = x_i \bmod m$, and embed the residue as a point on the unit circle:

$$\phi_m(r) = \big[\sin\big(\tfrac{2\pi r}{m}\big)\,,\ \cos\big(\tfrac{2\pi r}{m}\big)\big] \in \mathbb{R}^2.$$

Concatenating all 100 moduli gives $\mathbf{f}_i^{\text{mod}} \in \mathbb{R}^{200}$. This Sin/Cos embedding is equivariant to the group structure of $\mathbb{Z}/m\mathbb{Z}$ and avoids discontinuities at wrap-around boundaries.

3.3 Dual-Stream Embedding

The two feature vectors are projected to the model hidden dimension d by independent projection layers. A two-layer MLP is used for the Magnitude side:

$$\mathbf{h}_i^{\text{mag}} = \text{MLP}_{\text{mag}}(\mathbf{f}_i^{\text{mag}})\,, \quad \mathbf{h}_i^{\text{mod}} = W_{\text{mod}}\,\mathbf{f}_i^{\text{mod}} + \mathbf{b}_{\text{mod}}\,, \quad \mathbf{h}_i^{\text{mag}},\, \mathbf{h}_i^{\text{mod}} \in \mathbb{R}^d.$$

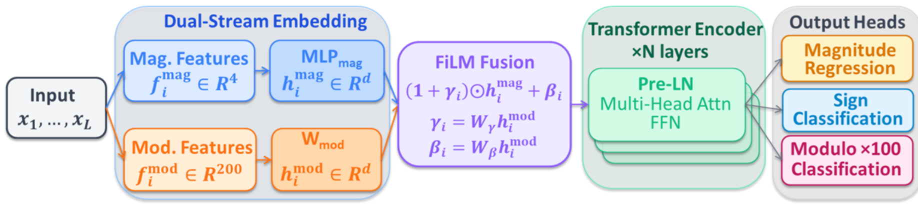

Fig. 1. IntSeqBERT architecture. The Dual-Stream Embedding block projects $\mathbf{f}_i^{\mathrm{mag}} \in \mathbb{R}^4$ and $\mathbf{f}_i^{\mathrm{mod}} \in \mathbb{R}^{200}$ to $\mathbb{R}^d$ via $\mathrm{MLP}_{\mathrm{mag}}$ and W_{mod}, then fuses them with FiLM: $\mathbf{e}_i = (1+\boldsymbol{\gamma}_i) \odot \mathbf{h}_i^{\mathrm{mag}} + \boldsymbol{\beta}_i$. Positional encodings are added before the Pre-LN Transformer encoder. Three prediction heads produce $\hat{v}, \hat{\sigma}^2 \in \mathbb{R}$ (magnitude regression), $\hat{s} \in \{+,-,0\}$ (sign classification), and $\hat{r}^{(m)} \in \{0,\dots,m-1\}$ (100× Modulo classification).

Table 1. Model configurations.

Config	Layers	d	Heads	Parameters
Small	6	256	4	6.4M
Middle	8	512	8	29.0M
Large	12	768	12	91.5M

3.4 FiLM Fusion

The two streams are fused by Feature-wise Linear Modulation (FiLM) [10]. The Modulo embedding generates element-wise scale $\boldsymbol{\gamma}_i$ and shift $\boldsymbol{\beta}_i$ to modulate the Magnitude embedding:

$$\boldsymbol{\gamma}_i = W_\gamma \, \mathbf{h}_i^{\mathrm{mod}}, \quad \boldsymbol{\beta}_i = W_\beta \, \mathbf{h}_i^{\mathrm{mod}},$$

$$\mathbf{e}_i = (1+\boldsymbol{\gamma}_i) \odot \mathbf{h}_i^{\mathrm{mag}} + \boldsymbol{\beta}_i.$$

A ReLU is applied after the Modulo projection, and dropout is inserted before FiLM. This formulation allows arithmetic periodicity to condition continuous-value Magnitude representations with parameter efficiency ($W_\gamma, W_\beta \in \mathbb{R}^{d\times d}$). Standard sinusoidal positional encodings are added to $\mathbf{e}_i$ before the encoder.

The overall architecture is illustrated in Fig. 1: each element is projected independently via the Magnitude stream (blue) and the Modulo stream (orange), fused by FiLM, and passed to the Transformer encoder.

3.5 Transformer Encoder

The fused sequence $(\mathbf{e}_1, \dots, \mathbf{e}_L)$ is processed by a standard Transformer encoder [16] with Pre-Layer Normalisation [17]. Three model sizes are evaluated:

3.6 Prediction Heads

Let $\mathbf{z}_i \in \mathbb{R}^d$ be the encoder output at masked position i.

Magnitude head (regression): A two-layer MLP ($d \to d \to 2$, ReLU activation) outputs $(\mu_i, \log \sigma_i^2)$; the prediction is $\hat{v}_i = \mu_i$.

Sign head (3-class classification): $\hat{s}_i = \mathrm{softmax}(W_{\mathrm{sign}}\, \mathbf{z}_i)$, $W_{\mathrm{sign}} \in \mathbb{R}^{3\times d}$.

Modulo head (100 independent classifiers): For each $m \in \{2, \dots, 101\}$, a linear layer outputs logits over $\{0, \dots, m-1\}$ sharing the same input $\mathbf{z}_i$ but with independent parameters. The total output dimension is $\sum_{m=2}^{101} m = 5{,}150$.

3.7 Training Objective

The multi-task loss is $\mathcal{L} = \mathcal{L}_{\mathrm{mag}} + \mathcal{L}_{\mathrm{sign}} + 2\,\mathcal{L}_{\mathrm{mod}}$ (weights fixed; adaptive weighting caused instability). $\mathcal{L}_{\mathrm{mag}}$ is Huber loss; $\mathcal{L}_{\mathrm{sign}}$ is cross-entropy over three sign classes; $\mathcal{L}_{\mathrm{mod}}$ is the mean cross-entropy of the 100 Modulo heads, normalised by $\log m$: $\mathcal{L}_{\mathrm{mod}} = \frac{1}{100}\sum_{m=2}^{101} \frac{1}{\log m}\mathcal{L}_{\mathrm{CE}}^{(m)}$. All losses are computed only at masked positions.

3.8 Baselines

Vanilla Transformer maps each integer to a token ID in a vocabulary of 20,003 entries (values 0–19,999 plus `PAD`, `MASK`, `UNK`); out-of-vocabulary values are replaced by `UNK`. The same three prediction heads are applied to the token embeddings. The vocabulary size 20,003 was chosen so that memory consumption matches IntSeqBERT under the 8 GB VRAM constraint. The prior benchmark FACT [1] handles values up to several million, presupposing larger compute resources than our setting.

Ablation (Magnitude-only) is identical to IntSeqBERT but uses only the Magnitude stream; the FiLM module is removed and $\mathbf{e}_i = \mathbf{h}_i^{\mathrm{mag}}$. This isolates the contribution of the Modulo stream.

3.9 IntegerSolver

The pre-trained model outputs Magnitude $(\mu_i, \log \sigma_i^2)$, sign, and Modulo distributions at masked positions. **IntegerSolver** recovers concrete integers from these predictions.

The Solver derives the 3σ interval $[n_{\min}, n_{\max}]$ from the Magnitude prediction and dynamically selects one of three modes based on the search width $\Delta n = |n_{\max} - n_{\min}|$:

When the sign prediction is zero, the Solver immediately returns 0. If no valid candidate is found in the search range, the call is recorded as `none`.

Each candidate n is scored by $\mathrm{score}(n) = -(v_n - \mu_i)^2/(2\sigma_i^2) + 0.3\sum_m \log P(n \bmod m)$; the coefficient 0.3 suppresses score inflation from moduli sharing prime factors. The top k candidates are returned and evaluated as Solver Top-k (Section 5.4).

Table 2. Solver operating modes.

Mode	Condition	Method
Dense	$\Delta n \leq 10^6$	Enumerate all integers in range
Sieve	$10^6 < \Delta n \leq 10^{14}$	CRT beam search anchored on high-confidence moduli
CRT	$\Delta n > 10^{14}$	Sparse CRT beam search to generate large integers directly

Table 3. Magnitude buckets used for scale-stratified evaluation.

Bucket	Range of u	Typical value range	Element fraction
Small	$u < 2$	$\lvert x\rvert < 10^2$	52.7%
Medium	$2 \leq u < 5$	$10^2 \leq \lvert x\rvert < 10^5$	29.9%
Large	$5 \leq u < 20$	$10^5 \leq \lvert x\rvert < 10^{20}$	16.2%
Huge	$20 \leq u < 50$	$10^{20} \leq \lvert x\rvert < 10^{50}$	1.1%
Astronomical	$u \geq 50$	$\lvert x\rvert \geq 10^{50}$	0.0%

4 Dataset and Experimental Setup

4.1 Dataset

As of January 2026, OEIS contains 391,710 sequences. We exclude sequences with fewer than ten terms and those carrying any of eleven OEIS tags incompatible with integer-sequence prediction (`cons`/`cofr`/`frac`/`base`/`word` for out-of-scope representations; `fini`/`tabl` for undefined task; `dead`/`unkn`/`less`/`dumb` for quality), yielding 274,705 sequences (70.1%) as our standard dataset (`std`).

The corpus is split 8:1:1 (seed 42): 219,765 training / 27,470 validation / 27,470 test sequences.

Each sample corresponds to a sequence prefix of at most $L = 128$ terms (longer sequences are truncated). In the test set, sequence lengths range from 10 to 168 terms (mean 42.5), ensuring that the Solver always receives at least 9 terms of preceding context (Section 5.4).

For scale-stratified evaluation, we define five Magnitude buckets based on $u = \log_{10} |x|$ ($u = 0$ when $x = 0$):

Limitations of the split. The random split may place related sequences (e.g., Fibonacci and Lucas) in different splits; family-aware splits are left for future work.

4.2 Training Configuration

All models share the following hyperparameters:

Some OEIS values reach $|x| \approx 10^{210}$, causing FP16 overflow; all computation is therefore conducted in FP32. All results are reported by applying the model trained for the fixed 200 epochs to the test set.

Table 4. Training hyperparameters (common to all models).

Hyperparameter	Value
Epochs	200 (no early stopping)
Batch size	32 (gradient accumulation 2 steps; effective 64)
Learning rate	5×10^{-5}
Warmup fraction	10%
Optimiser	AdamW, weight decay 0.01
Numerical precision	FP32 (AMP disabled)
Mask probability	0.15
GPU	GeForce RTX 3070 Ti (8 GB VRAM) $\times$ 1

4.3 Evaluation Metrics

Magnitude accuracy ($\mathrm{Acc}_{0.5}$): fraction of masked positions with $|\hat{v}_i - v_i| < 0.5$ ($\sqrt{10}{\approx}3.16\times$ tolerance in integer scale). **Sign accuracy**: fraction with correct predicted sign class. **MMA (Mean Modulo Accuracy)**: mean classification accuracy across all 100 moduli. **Solver Top-k**: fraction of next-term instances where the true value appears in the Solver's top-k candidates (Section 5.4). **NIG (Normalised Information Gain)**: $1 - \mathcal{L}_{\mathrm{CE}}^{(m)} / \ln m$ for modulus m; higher NIG indicates stronger learned periodic structure.

5 Experiments

5.1 Main Results

Table 5 shows test-set performance for all model sizes and variants. IntSeqBERT consistently outperforms both baselines on all scales and metrics. At the Large scale, IntSeqBERT surpasses the Vanilla baseline by +8.9 pt in Mag Acc and +4.5 pt in MMA. The Ablation model shows the largest MMA drop at Large scale (−15.2 pt), directly quantifying the contribution of the arithmetic-periodicity features. Notably, the Ablation maintains competitive Mag Acc, indicating that sign and Modulo information contribute only marginally to magnitude regression but are essential for Modulo prediction.

Learning curves. Fig. 2 shows validation loss across all variants. Large IntSeqBERT converges from 2.17 to 1.01 over 200 epochs with no overfitting (Train 1.00 / Val 1.01). Vanilla (1.77) and Ablation (1.39) maintain higher final loss.

5.2 Magnitude Prediction

Scale-stratified analysis. Table 6 reports per-bucket MSE on the test set for Large models. The Vanilla model suffers catastrophic degradation in the Large bucket (MSE = 2.10, 13× that of IntSeqBERT), caused by all out-of-vocabulary integers being absorbed into the `UNK` token. IntSeqBERT achieves the best MSE in all buckets except Small, with particularly pronounced advantage at Medium

Table 5. Test results. Mag Acc = $\text{Accuracy}_{0.5}$ (%), Sign Acc (%), MMA = Mean Modulo Accuracy (%). **Bold** indicates the best value within each size group.

Size	Model	Mag Acc	Sign Acc	MMA
Small	**IntSeq**	**94.73**	**97.78**	**40.43**
Small	Vanilla	85.73	96.91	36.21
Small	Ablation	93.72	97.39	25.97
Middle	**IntSeq**	**95.71**	**98.34**	**46.88**
Middle	Vanilla	87.37	97.42	42.53
Middle	Ablation	92.45	97.90	31.93
Large	**IntSeq**	**95.85**	**98.54**	**50.38**
Large	Vanilla	86.97	97.66	45.85
Large	Ablation	89.70	98.29	35.22

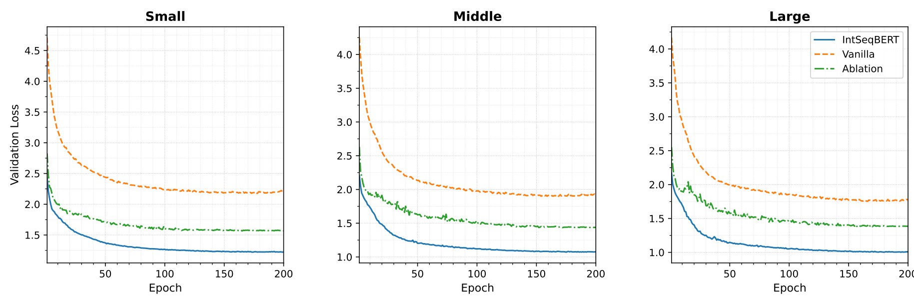

Fig. 2. Validation loss learning curves for all scales (Small / Middle / Large) and all variants. IntSeqBERT (solid blue) consistently achieves lower loss than Vanilla (dashed orange) and Ablation (dash-dot green). At the Large scale, IntSeqBERT converges to Val Loss = 1.01 at epoch 200.

and above. The Ablation degrades in the Medium bucket without Modulo context (MSE = 0.116 vs. IntSeqBERT 0.051), and collapses more severely in the Huge and Astronomical buckets, indicating that FiLM modulation by the Modulo stream acts as an arithmetic structural constraint on Magnitude estimation for large integers.

Fig. 3 shows predicted vs. true Magnitude scatter plots for the three Large model variants. IntSeqBERT achieves the highest coefficient of determination (R^2 = 0.988), substantially reducing off-diagonal dispersion in the Large and Huge buckets compared to Vanilla (R^2 = 0.943), visually corroborating the MSE results of Table 6.

Calibration. Fig. 4 shows uncertainty calibration for Large models. Vanilla is severely miscalibrated (ECE = 5.36), with overconfidence at low scale and exploding σ at high scale. IntSeqBERT (ECE = 0.65) matches Ablation (ECE = 0.66), indicating calibration is driven by continuous representation rather than the Modulo stream.

Table 6. Scale-stratified MSE on the test set (Large models). Lower is better.

Bucket	IntSeq	Vanilla	Ablation
Small ($u < 2$)	0.111	0.138	**0.103**
Medium ($2 \le u < 5$)	**0.051**	0.071	0.116
Large ($5 \le u < 20$)	**0.162**	2.100	0.381
Huge ($20 \le u < 50$)	**2.082**	22.73	5.021
Astronomical ($u \ge 50$)	**110.4**	840.0	532.6

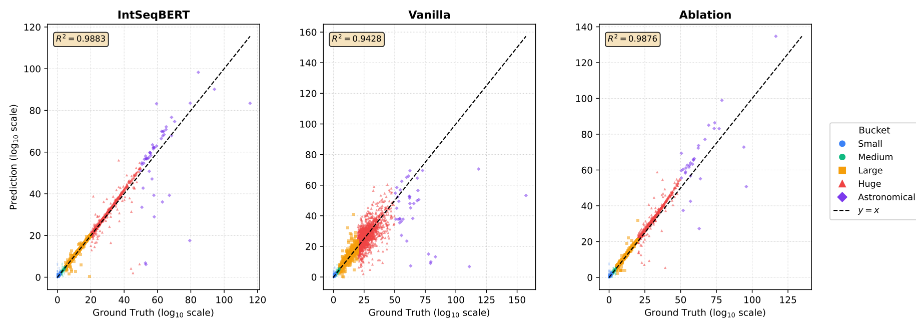

Fig. 3. Predicted vs. true Magnitude ($\log_{10}$ scale) for Large models. Points are coloured by bucket (Small=blue circle, Medium=green circle, Large=yellow-orange square, Huge=red triangle, Astronomical=purple diamond). IntSeqBERT achieves $R^2 = 0.988$ vs. Vanilla $R^2 = 0.943$; Vanilla shows pronounced scatter above the Large bucket.

5.3 Modulo Spectrum Analysis

Fig. 5 shows the NIG spectrum for $m = 2, \ldots, 101$ using Large models. IntSeqBERT (solid blue) exceeds both Vanilla and Ablation across the entire spectrum, with a visually clear contrast between prime and composite moduli.

Finding 1: NIG is strongly negatively correlated with Euler's totient ratio. A Pearson correlation of $r = -0.851$ ($p < 10^{-28}$) is observed between NIG and $\varphi(m)/m = \prod_{p|m}(1-1/p)$ (Fig. 6). Composite moduli with many small prime factors achieve higher NIG, consistent with a CRT aggregation effect: when m is a common multiple of smaller moduli $m_1, m_2, \ldots$, then $x \bmod m$ encodes the information of all those moduli simultaneously. The highest NIG across all models and scales is achieved at $m = 96 = 2^5 \times 3$ ($\varphi(96)/96 = 1/3$; Large IntSeq: NIG = 0.629, 95% CI [0.622, 0.634]), consistent with $m = 96$ being the largest modulus with totient ratio 1/3 among $\{12, 24, 48, 72, 96\}$ and thus distinguishing the widest value range. The exception is $m = 2$ (prime, NIG = 0.628, second highest), reflecting the corpus-wide ubiquity of parity in OEIS sequences.

Finding 2: Parity (mod 2) accuracy stratifies models. Mod-2 accuracy at the Large scale is 85.65% (IntSeq), 81.40% (Vanilla), and 72.13% (Ablation), making parity the single modulus most sensitive to the Modulo stream (-13.5 pt when removed). Table 7 lists representative modulus accuracies.

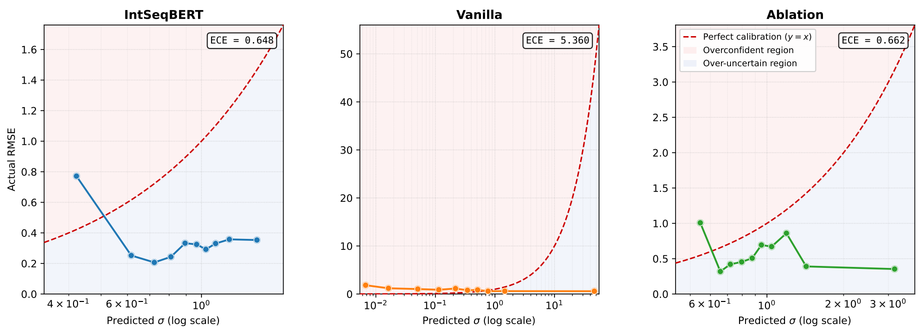

Fig. 4. Uncertainty calibration curves for Large models (x-axis: predicted σ (log scale), y-axis: actual RMSE). The red region (above diagonal) indicates overconfidence; blue indicates overestimation. IntSeqBERT (ECE = 0.648) and Ablation (ECE = 0.662) lie close to the diagonal, while Vanilla (ECE = 5.360) shows severe overconfidence at low σ.

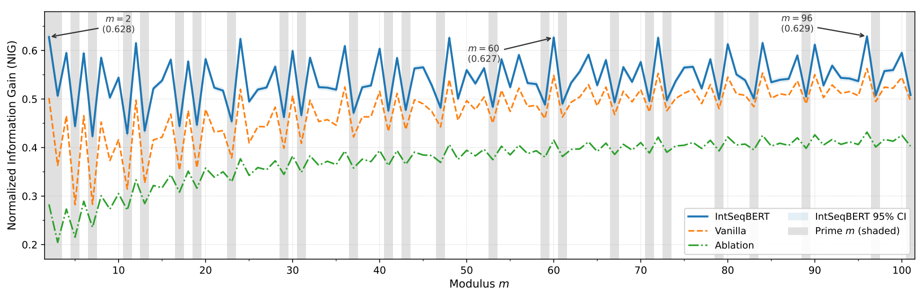

Fig. 5. NIG spectrum for moduli $m = 2, \ldots, 101$ (Large models). Grey shading marks prime moduli. The 95% CI for IntSeqBERT (light blue band) is computed by bootstrapping.

5.4 Solver-Based Next-Term Prediction

The Solver reconstructs candidate integers from the model's Magnitude, sign, and Modulo predictions and ranks them by likelihood. We evaluate exact-match accuracy on 10,000 test samples, where each sample corresponds to one OEIS sequence and the target is the *last term* (all preceding terms are given as context). Sequence lengths in the test set range from 10 to 168 (median 36, mean 42.5), so the Solver always receives at least 9 terms of context.

Valid-candidate rate is the fraction of samples for which the Solver returns at least one candidate. Vanilla always returns a candidate from its vocabulary softmax (rate = 100%). IntSeqBERT and Ablation may fail to find a valid candidate within the search range (`none` mode; 13.4% of Large-IntSeq calls), reducing their valid-candidate rate.

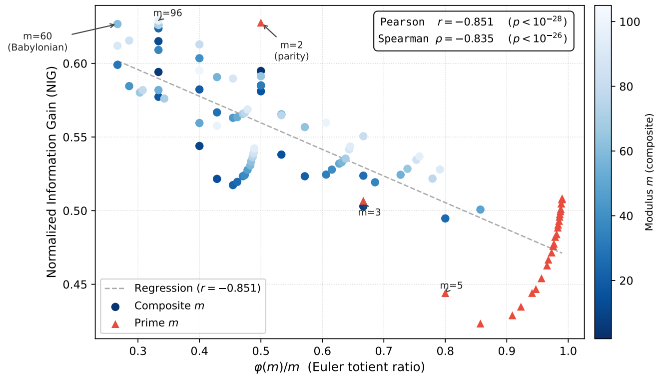

Fig. 6. NIG vs. Euler's totient ratio $\varphi(m)/m$ (Large IntSeqBERT). Composite moduli (blue circles, shade proportional to m) and prime moduli (red triangles) are shown separately. The regression line (grey dashed) indicates Pearson $r = -0.851$ ($p < 10^{-28}$). Notable moduli $m = 2$ (parity), $m = 60$ (Babylonian number), and $m = 96$ (highly composite) are annotated.

Table 7. Representative modulus accuracies for Large models (%).

m	IntSeq	Vanilla	Ablation	Interpretation
2	85.65	81.40	72.13	Parity
3	72.62	65.22	53.72	Ternary residue
5	60.37	50.07	42.63	Last decimal digit parity
10	58.38	49.25	39.47	Least significant decimal digit
60	53.97	47.87	35.12	Babylonian number (highly composite)
96	51.82	47.29	34.44	Highly composite
100	48.51	45.60	33.51	Centesimal residue

Table 8 shows Solver evaluation results. Large-scale IntSeqBERT achieves a Top-1 accuracy of 19.09%, a **7.4**× improvement over the Vanilla baseline (2.59%).

Accuracy by Magnitude bucket. Table 9 compares Solver Top-1 accuracy across buckets for all three Large models. IntSeqBERT achieves 68.34% Top-1 in the Small bucket, while the Ablation reaches 54.55% and Vanilla only 14.11%. Vanilla collapses to 0% for Medium and above due to the UNK-token degradation of its Magnitude predictions, whereas IntSeqBERT retains meaningful accuracy up to the Medium bucket (20.82%).

Mode breakdown (Large IntSeqBERT). Dense mode (24.0% of calls) achieves the highest Top-1 accuracy at 61.06%; Sieve mode (36.7%) achieves 5.36%; the zero-prediction shortcut (2.7%) achieves 89.96%. CRT mode (23.2%) achieves near-zero accuracy (0.09%); 13.4% of calls return no valid candidate. The CRT limitation is discussed in Section 6.3.

Table 8. Solver evaluation: Top-1 and Top-10 exact-match accuracy (%) and valid-candidate rate.

Size	Model	Top-1	Top-10	Sign Acc	Valid rate (%)
Small	**IntSeq**	**14.05**	**21.00**	**98.73**	90.59
Small	Vanilla	2.43	3.24	92.92	100.0
Small	Ablation	7.42	17.33	98.50	90.17
Middle	**IntSeq**	**17.02**	**22.62**	**99.02**	86.31
Middle	Vanilla	2.43	3.41	92.71	100.0
Middle	Ablation	9.88	20.52	98.74	90.34
Large	**IntSeq**	**19.09**	**26.23**	**99.02**	86.64
Large	Vanilla	2.59	3.80	92.05	100.0
Large	Ablation	11.75	21.79	98.94	86.99

Table 9. Solver Top-1 / Top-10 accuracy (%) by Magnitude bucket (Large models). **Bold**: best per bucket.

	IntSeqBERT		Vanilla		Ablation	
Bucket	Top-1	Top-10	Top-1	Top-10	Top-1	Top-10
Small	**68.34**	**88.50**	14.11	20.71	54.55	86.92
Medium	**20.82**	**31.50**	0.00	0.00	5.61	18.88
Large	**0.31**	**0.67**	0.00	0.00	0.03	0.05
Huge	**0.09**	**0.18**	0.00	0.00	0.00	0.00
Astronomical	0.00	0.00	0.00	0.00	0.00	0.00

6 Analysis and Discussion

6.1 Ablation Study: Contribution of the Modulo Stream

To isolate the contribution of the Modulo stream and FiLM fusion, we compare IntSeqBERT with the Magnitude-only Ablation model. As seen in Table 5, the Modulo stream delivers the largest gains in Modulo prediction (MMA +15.2 pt, parity +13.5 pt at Large scale). Its contribution to Magnitude prediction is smaller but significant ($\mathrm{Acc}_{0.5}$ +6.2 pt, MSE −0.228), consistent with the intuition that knowing $x_i \bmod m$ for $m \in \{2, \ldots, 101\}$ substantially constrains the set of plausible Magnitude values. Solver accuracy improves by +7.3 pt, primarily because more accurate residue information refines scoring in Dense and Sieve modes.

6.2 Scaling Behaviour

Table 10 shows how IntSeqBERT's metrics improve with model size. Magnitude accuracy improves only modestly from Small to Large (+1.1 pt), whereas Modulo accuracy improves substantially (+10.0 pt). This is consistent with the intuition that modular arithmetic is more compositional and thus benefits more from increased representational capacity. Solver Top-1 accuracy follows the Modulo trend with consistent improvement (+5.0 pt).

Table 10. IntSeqBERT scaling: test-split metrics (Small / Middle / Large).

Metric	Small	Middle	Large	Δ (S→L)
Mag Acc (%)	94.73	95.71	95.85	+1.12
MMA (%)	40.43	46.88	50.38	+9.95
mod-2 accuracy (%)	81.97	84.50	85.65	+3.68
Solver Top-1 (%)	14.05	17.02	19.09	+5.04
Magnitude MSE	0.228	0.164	0.142	−0.086

The trends in Table 10 are visualised in Fig. 7, which contrasts the differential scaling of Modulo and Magnitude accuracy across model sizes.

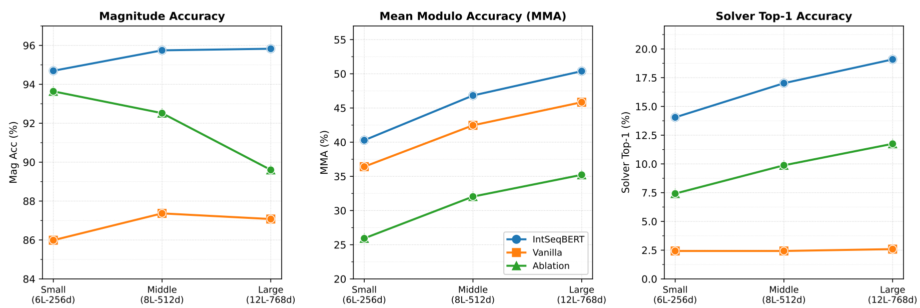

Fig. 7. Scaling behaviour (Small / Middle / Large). Left: Mag Acc improves only +1.1 pt. Centre: MMA for IntSeqBERT improves +10.1 pt (40.3%→50.4%). Right: Solver Top-1 improves +5.0 pt, with a growing gap over Vanilla.

6.3 Limitations

Difficulty with large integers. Solver accuracy collapses to near zero for $|x| \geq 10^{20}$ (Huge and Astronomical buckets): CRT-based reconstruction requires *exact* residues for multiple moduli simultaneously, and errors are frequent enough at MMA≈50% to cause failure (Top-1 = 0.09% in CRT mode). Approximately 13% of IntSeqBERT Solver calls return no valid candidate (`none` mode), primarily due to CRT failures; approximate CRT or relaxed residue constraints are a natural remedy.

Dataset bias. OEIS sequences are predominantly non-negative (`nonn`). The high sign accuracy (98.54%) may partially reflect this bias. Astronomical-bucket metrics are based on few samples and should be interpreted with caution.

7 Conclusion

We have presented **IntSeqBERT**, a dual-stream Transformer encoder for integer sequences that fuses a continuous log-scale Magnitude embedding with

Sin/Cos Modulo embeddings via FiLM, and is jointly trained on 219,765 OEIS sequences. Experiments (Section 5) demonstrate that the dual-stream design substantially outperforms both a standard tokenised Transformer and a Magnitude-only ablation across all scales and metrics, with especially large gains in Modulo accuracy and Solver-based next-term prediction.

Future work.

- Combining Magnitude uncertainty estimation with approximate CRT to improve large-integer prediction, and extending the Modulo stream beyond $m = 101$.
- Constructing family-aware dataset splits and introducing sequence-synthesis data augmentation [1] to improve generalisation in sparse Magnitude buckets.
- Scaling to larger models and pre-training on the full OEIS corpus.

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